17 January 2024 09:01

EE 2004 Digital Signal Processing.

Objective : Introduce fundamentals of Discrete-time Signals and Systems

Syllabus :

- Review of Discrete time Signals / Systems (DT)

Analytical Tools
- Discrete time Fourier Transform
- Z-Transform

Practical Applications
- Frequency domain Analysis (Digital Filtering) — Basics
- Sampling
- Discrete Fourier Transform (DFT)

Textbook : Discrete-time signal Processing by Oppenheim & Schafer

Evaluation :

Quiz	1	20 %
	2	20 %.
Final Exam		60 %

Discrete-time Signals (DT)

DT signal $\{x(n)\}$ is a function defined over discrete-time variable "n".

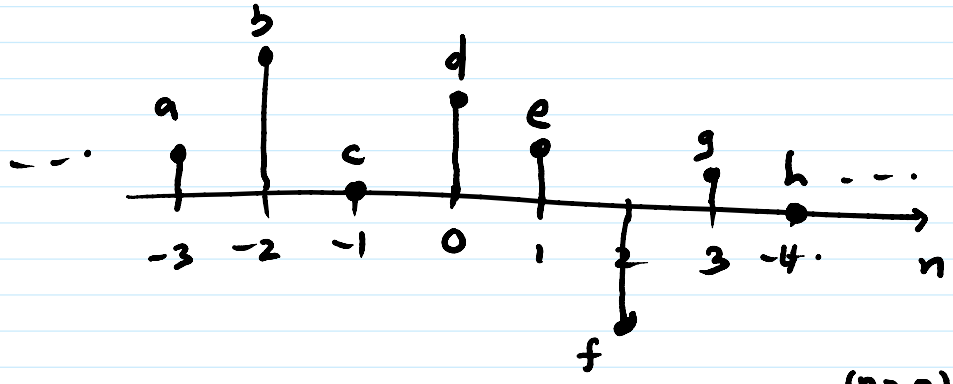
In our course, "n" takes only integer values.

$$\{x(n)\} : \mathbb{Z} \longrightarrow \mathbb{R} \text{ (real)} \quad \text{or} \quad \mathbb{C} \text{ (complex)}$$

$\{x(n)\} \equiv$ A sequence of real or complex numbers indexed by "n".

Example $\{x(n)\}$

$$x(n) = \begin{cases} a; & n=-3 \\ b; & n=-2 \\ \vdots \end{cases}$$



$$\{x(n)\} = \{\ldots, a, b, c, \underset{\downarrow}{d}, e, f, g, h, \ldots\} \quad (n=0)$$

Notation: $\{x(n)\}$ → denotes entire signal/sequence

$x(n) \Rightarrow$ refers to the value of signal x at time index n.

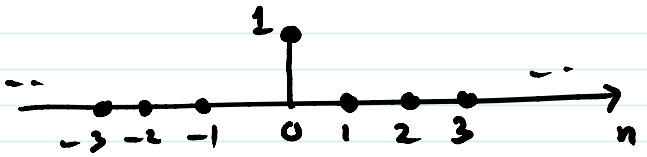
$x(-1) = c$

Basic DT Signals.

1. Unit impulse signal

$$\delta(n) = \begin{cases} 1 ; & n = 0 \\ 0 ; & n \neq 0 \end{cases}$$

Support $[0, 0]$



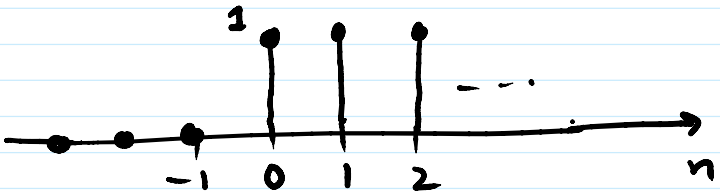
$$\{\delta(n)\} = \{ \cdots, 0, \underset{\uparrow}{1}, 0, \cdots \}$$

2. Unit step Signal

$$u(n) = \begin{cases} 1 ; & n \geq 0 \\ 0 ; & n < 0 \end{cases}$$

Support $[0, \infty)$

len $= \infty$



Signal ATTRIBUTES

- Support is the smallest (single) interval $[N_1, N_2]$ of $\{x(n)\}$

- Support of $\{x(n)\}$ is the smallest interval $[N_1, N_2]$ which contains all non-zero entries of $\{x(n)\}$
- Length of signal $\{x(n)\}$ with support $[N_1, N_2]$ is defined $N_2 - N_1 + 1$

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3. Real Cosine Signal

$$x(n) = A\cos(\omega_0 n + \phi), \quad n \in \mathbb{Z}$$

support $(-\infty, \infty)$

$$-\infty < n < \infty$$

Phase $\phi \in [-\pi, \pi]$

Amplitude $A \geq 0$

$$-\cos(t) = \cos(t+\pi)$$

$$\cos(-t) = \cos(t)$$

$\omega_0 \in [0, \pi]$ Frequency $\omega_0 \geq 0$

$$\cos((\omega_0 + 2\pi)n)$$

$$= \cos(\omega_0 n + 2\pi n)$$

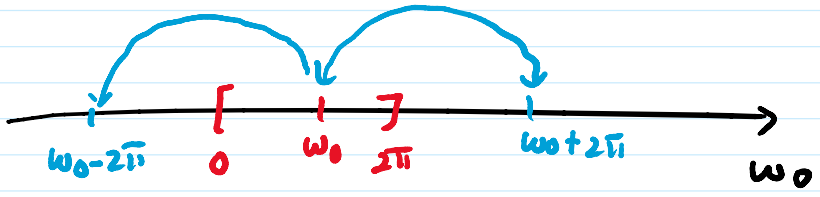
$$= \cos(\omega_0 n)$$

$\Downarrow$ Restrict

$\omega_0 \in [0, 2\pi]$

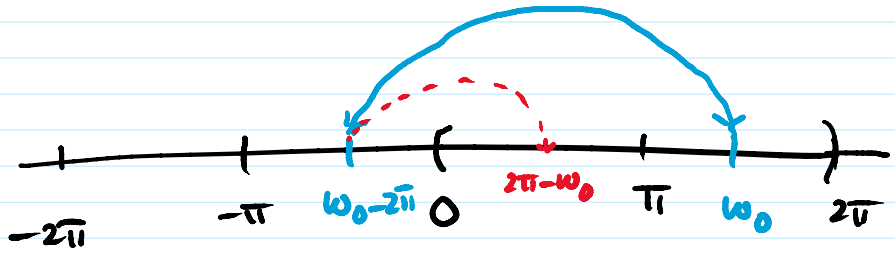
$\Downarrow$ Restrict

$\omega_0 \in [0, \pi]$



$\omega_0 \equiv \omega_0 + 2\pi k$ for any integer k.

integer k.



Suppose $\omega_0 \in [\pi, 2\pi]$ $\quad \cos(\omega_0 n)$

$\Downarrow$ $\quad\quad\quad$ $\|$

$2\pi - \omega_0 \in [0, \pi]$ $\quad \cos((2\pi - \omega_0)n)$

DT Cosine Signal

ω_0 close to 0 $\Rightarrow$ Low freq.

ω_0 close to π $\Rightarrow$ High freq.

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4. Complex Exponential Signal $\quad j = \sqrt{-1}$

$$x(n) = A e^{j(\omega_0 n + \phi)}, \quad -\infty < n < \infty$$

$$= A\cos(\omega_0 n + \phi) + jA\sin(\omega_0 n + \phi)$$

$A \geq 0$

$\phi \in (-\pi, \pi)$

$\omega_0 \in (-\pi, \pi]$ or $[0, 2\pi]$

$\omega_0 \in [-\pi, \pi]$ or $[0, 2\pi]$

$\Downarrow$

$[-\pi, 0] \cup [0, \pi]$ $\qquad$ $[0, \pi] \cup [\pi, 2\pi]$

$+2\pi$

$$e^{j(\omega_0 + 2\pi)n} = e^{j\omega_0 n} \cdot \underbrace{e^{j2\pi n}}_{1}$$

For complex expmentials

$$e^{+j\omega_0 n} \neq e^{-j\omega_0 n}$$

$+\omega_0$ & $-\omega_0$ are different.

Example:

$x(n) = e^{j(0.5)n}$ $\qquad$ $e^{-j0.5n} = y(n)$

